

CSE331 Assignment 2

Tanhia Fatema Afnan [TNF]

- The assignment needs to be submitted online in pdf format in a google form that will be published later.
- Deadline for entire Assignment 2 submission:
7 January, 2026, Wednesday, 11:59 pm.
- Only one submission per group.
- The submitted document should have clear and legible scans.

1. (20 marks) Prove that the following languages are nonregular using the Pumping lemma:

- a. $L = \{ w \in \{0, 1\}^* \mid 0^i 1^j 0^k, i + k > j, i, j, k > 0 \}$
- b. $L = \{ a^n b^l a^k \mid k > n + l \}$
- c. $L = \{ 0^n l \mid n \geq 0 \}$
- d. $L = \{ a^n b^l \mid n \neq l \}$
- e. $L = \{ w \in \{0, 1\}^* \mid w \text{ is a palindrome} \}$

2. (30 marks) Write CFG for the following languages:

- a. $L = \{ a^n b^m \mid m, n > 0, n \neq m \}$
- b. $L = \{ a^n b^m \mid m, n > 0, n = 4m + 2 \}$
- c. $L = \{ a^n b^m c^p \mid m, n, p > 0, m=n \text{ or } m=p \}$
- d. $L = \{ w_1 \# w_2 \mid \text{the number of 01 in } w_1 \text{ is equal to the number of 10 in } w_2 \}$
- e. $L = \{ w_1 \# w_2 \mid w_1 \text{ contains 0 in every other 3rd position, } |w_2| = 2|w_1| \}$
- f. $L = \{ w_1 \# w_2 \mid |w_1| \text{ is odd and } |w_2| \text{ is even, } w_1 \text{ contains the substring 100} \}$
- g. $L = \{ w = \{a, b\}^* \mid \text{length is a multiple of 3, the first third part contains at least 2 b's, the second third part contains only a} \}$

3. (10 marks) Answer from the following grammar:

$S \rightarrow PE$
 $P \rightarrow aPa \mid bPb \mid a \mid b$
 $E \rightarrow aaE \mid abE \mid baE \mid bbE \mid \varepsilon$

- a. Give a leftmost derivation for the string **bababba**.
- b. Sketch the parse tree corresponding to the derivation you gave in (a).
- c. Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string.
- d. Find how many parse trees the string **aababab** has in the given grammar.

4. (20 marks) Draw Push-down Automata for the following languages:

- a. $L = \{ 0^i 1^j 0^k \mid i + j < k \}$
- b. Q. 2 (a)
- c. Q. 2 (b)
- d. Q. 2 (d)
- e. Q. 2 (g)

5. (20 marks) Draw Turing Machine for the following languages:
- a. Given an input string w over the alphabet $\{a, b\}$, writes the same string again after a delimiter symbol $\#$. The TM should halt with the tape containing $w\#w$, where the first occurrence of w is the input and the second is its copy.
 - b. $L = \{w \in \{0, 1\}^* \mid w \text{ is a palindrome}\}$
 - c. $L = \{a^i b^j c^k \mid i = j = k\}$
 - d. Add two binary numbers. The tape initially contains $w_1 c w_2$. w_1 and w_2 are binary numbers with LSB in the leftmost bit and c as separator. Write the sum to the right side of the second c .